

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Ebin Antony P (192373011)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

APPLICATION- BASED QUESTIONS

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Given Ciphertext

GSRH RH Z HVXIVG

The cybersecurity intern has intercepted this message during network monitoring. The message appears to use a **simple substitution cipher** rather than a complex encryption algorithm.

a. Deduce the most likely original message

The most likely original message is:

THIS IS A SECRET

The ciphertext can be decoded using the **Atbash cipher**.

What is the Atbash Cipher?

The Atbash cipher is a classical **substitution cipher**. It replaces every letter of the alphabet with the letter in the reverse position.

The alphabet is arranged as follows:

A = Z	N = M
B = Y	O = L
C = X	P = K
D = W	Q = J
E = V	R = I
F = U	S = H
G = T	T = G
H = S	U = F
I = R	V = E
J = Q	W = D
K = P	X = C
L = O	Y = B
M = N	Z = A

Since Atbash is a reciprocal substitution, the same table can be used for decryption.

Step-by-step decryption

The ciphertext is:

GSRH RH Z HVXIVG

Separate the words:

GSRH | RH | Z | HVXIVG

Now decode each letter.

Word 1: GSRH

Cipher letter	Original letter
G	T
S	H
R	I
H	S

Therefore:

GSRH → THIS

Word 2: RH

Cipher letter	Original letter
R	I
H	S

Therefore:

RH → IS

Word 3: Z

Cipher letter	Original letter
Z	A

Therefore:

Z → A

Word 4: HVXIVG

Cipher letter	Original letter
H	S
V	E
X	C
I	R
V	E
G	T

Therefore:

HVXIVG → SECRET

Combining all the words:

GSRH RH Z HVXIVG

becomes:

THIS IS A SECRET

b. Justification of the answer

The answer can be justified by examining the **word structure, repeated letters, and characteristics of classical substitution ciphers.**

1. Word-length analysis

The ciphertext contains four words:

GSRH RH Z HVXIVG

Their lengths are:

- GSRH → 4 letters
- RH → 2 letters
- Z → 1 letter
- HVXIVG → 6 letters

Therefore, the original message should also have the pattern:

4 letters – 2 letters – 1 letter – 6 letters

The phrase:

THIS IS A SECRET

has exactly the same pattern:

4 – 2 – 1 – 6

This strongly supports the solution.

2. Single-letter word

The ciphertext contains the single-letter word:

Z

In English, common one-letter words are generally:

- **A**
- **I**

Using the Atbash substitution:

Z → **A**

This gives the meaningful phrase:

THIS IS A SECRET

3. Repeated letters

The letter **H** appears in multiple locations.

For example:

GSRH RH Z HVXIVG

The H in GSRH and the H in RH decrypt to the same letter:

H → **S**

This is consistent with a substitution cipher because the same ciphertext letter normally represents the same plaintext letter.

4. Letter substitution pattern

Consider:

GSRH

The substitution gives:

G → **T**

S → **H**

R → I

H → S

So:

GSRH → THIS

This produces a common English word.

Similarly:

RH → IS

and:

HVXIVG → SECRET

The fact that **all four words produce meaningful English** is strong evidence that the decryption is correct.

5. Known characteristics of Atbash

Atbash is one of the simplest substitution ciphers. It does not use a changing key for each letter. Instead, every letter has a fixed opposite letter.

For example:

A ↔ Z

B ↔ Y

C ↔ X

This makes the cipher easy to recognize when the resulting plaintext forms meaningful English.

Conclusion for Question 2

The intercepted message:

GSRH RH Z HVXIVG

is most likely encrypted using the **Atbash substitution cipher**.

After applying the reverse-alphabet substitution:

GSRH RH Z HVXIVG → THIS IS A SECRET

The answer is supported by the matching word lengths, single-letter word, repeated-letter structure, and the fixed substitution characteristics of the Atbash cipher.

1. Atbash Cipher Mapping

Plaintext (A-Z)	A	B	C	D	E	F	G	H	I	J	K	L	M
Ciphertext (Z-A)	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

Plaintext (N-Z)	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext (Z-A)	M	L	K	J	I	H	G	F	E	D	C	B	A

2. Decryption Step-by-Step

Ciphertext: **GSRH RH Z HVXIVG**

GSRH → THIS

G → T
S → H
R → I
H → S

GSRH = THIS

RH → IS

R → I
H → S

RH = IS

Z → A

Z → A

Z = A

HVXIVG → SECRET

H → S
V → E
X → C
I → R
V → E
G → T

HVXIVG = SECRET

Original Message: **THIS IS A SECRET**

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as:

“WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Given Ciphertext

WECRLTEERDSOEFEAOCAIVDEN

The company uses a legacy **transposition-based encryption method with three levels of arrangement**.

This strongly suggests the use of a **3-rail Rail Fence cipher**.

a. Determine the most probable original message

What is a transposition cipher?

A transposition cipher is an encryption technique in which the **letters themselves are not replaced**.

Instead, the letters are simply **rearranged into a different order**.

For example, suppose the original message is:

HELLO

A transposition cipher might rearrange the letters as:

LHOEL

The same letters are present, but their positions have changed.

This is different from a substitution cipher such as Atbash, where letters themselves are replaced.

What is the Rail Fence cipher?

The Rail Fence cipher is a classical transposition cipher.

In a **3-rail Rail Fence cipher**, the plaintext is written diagonally across three rows.

For example:

Rail 1:	W		A		D		E
Rail 2:		E	R	S	O	V	R
Rail 3:			A		I		C

The letters are then read row by row to create the ciphertext.

Because the question states that the system uses **three levels of arrangement**, a 3-rail Rail Fence cipher is the most probable interpretation.

Given ciphertext

WECRLTEERDSOEFEAOCAIVDEN

The ciphertext contains **24 letters**.

The probable original message is:

WE ARE DISCOVERED FLEE AT ONCE

Removing spaces:

WEAREDISCOVEREDFLEEATONCE

Count the letters:

WEAREDISCOVEREDFLEEATONCE

This contains 24 letters, which matches the ciphertext.

Step-by-step arrangement

For a 3-rail Rail Fence cipher, the writing pattern follows:

Rail 1:	1		5		9		13		17		21													
Rail 2:		2		4		6		8		10		12		14		16		18		20		22		24
Rail 3:			3			7			11			15			19			23						

The plaintext:

WEAREDISCOVEREDFLEEATONCE

is placed in a zig-zag pattern.

The important idea is that the letters move:

Top → Middle → Bottom → Middle → Top → ...

This creates three rows or rails.

When the characters are subsequently read **row by row**, the resulting encrypted text is:

WECRLTEERDSOEFEAOCAIVDEN

Therefore, reversing the process gives:

WE ARE DISCOVERED FLEE AT ONCE

Meaning of the recovered message

The plaintext says:

WE ARE DISCOVERED FLEE AT ONCE

This is a meaningful English sentence and sounds like a confidential warning.

The message tells the receiver that the sender's group has been discovered and that they should leave immediately.

This also explains why such a message would need encryption in a legacy communication system.

Transposition Techniques

Rail Fence Cipher with 3 rails to encrypt the message

WE ARE DISCOVERED RUN AT ONCE

Step 1: Prepare the Text

First, remove the spaces to get the plaintext ready for encryption:

Plaintext: WEAREDISCOVEREDRUNATONCE

Step 2: Arrange the Text on Rails

Write the message in a zigzag pattern across 3 rails:

```
W . . . E . . . C . . . R . . . N . . . N . . . E
. E . R . D . S . O . E . E . R . U . A . O . C .
. . A . . . I . . . V . . . D . . . T . . . E . .
```

b. Critically evaluate why this method is vulnerable

Although the Rail Fence cipher was useful historically for hiding messages from casual readers, it is **not secure enough for modern corporate communication**.

There are several reasons.

1. It does not change the letters

The biggest weakness is that the cipher only **rearranges letters**.

For example, if the plaintext contains many occurrences of the letter E, the ciphertext will also contain exactly the same number of Es.

Nothing is actually removed or mathematically hidden.

Therefore, the statistical properties of the original message remain largely visible.

2. Very small key space

The security of a Rail Fence cipher depends mainly on the number of rails.

If the organization uses only three rails, an attacker does not have many possibilities to test.

For example, an attacker could try:

- 2 rails
- 3 rails
- 4 rails
- 5 rails
- 6 rails

and compare the resulting text.

With modern computers, trying these possibilities is extremely easy.

3. Vulnerable to brute-force experimentation

A brute-force attack involves trying different possible arrangements until meaningful plaintext is obtained.

Since the number of possible rail configurations is small, an attacker can automate this process.

For a short corporate message, a human attacker may even be able to identify the correct arrangement manually.

4. Letter frequency is preserved

In a substitution cipher, the letters are changed but their frequency can still sometimes be analyzed.

In a transposition cipher, the situation is even weaker because the actual letters remain unchanged.

For example, suppose the plaintext contains:

- 5 E's
- 3 A's
- 2 T's

The ciphertext will still contain:

- 5 E's
- 3 A's
- 2 T's

Only their positions are changed.

This provides useful information to an attacker.

5. No authentication

The Rail Fence cipher does not provide any method to verify **who actually sent the message**.

An attacker could potentially modify the encrypted message.

The receiver would have no cryptographic mechanism to determine whether the message was altered.

6. No integrity protection

Modern secure communication requires more than hiding information.

It should also ensure that the message has not been modified during transmission.

A simple Rail Fence cipher does not provide this protection.

For example, if an attacker changes or removes some encrypted characters, there is no built-in mechanism to detect the modification.

7. No protection against modern attacks

Classical ciphers such as Rail Fence were designed for a much older threat environment.

Modern attackers can use:

- Automated cryptanalysis
- High-speed computers
- Pattern analysis
- Brute-force techniques
- Network monitoring
- Automated scripts

Therefore, a simple 3-rail transposition cipher should not be considered secure today.

c. Practical enhancement

A practical improvement would be to **replace the legacy Rail Fence cipher with a modern authenticated encryption algorithm**, such as AES-GCM.

What is AES-GCM?

AES stands for:

Advanced Encryption Standard

GCM stands for:

Galois/Counter Mode

AES-GCM is widely used as a modern authenticated-encryption approach.

Instead of simply rearranging letters, AES performs strong mathematical transformations using a secret cryptographic key.

Advantages of AES-GCM

1. Strong confidentiality

The original message is transformed into ciphertext that cannot practically be understood without the correct key.

2. Larger security space

Modern cryptographic keys are far more difficult to guess than the small number of possible arrangements in a 3-rail cipher.

3. Integrity protection

AES-GCM can detect whether encrypted data has been modified.

If an attacker changes the ciphertext, authentication verification can fail.

4. Authentication

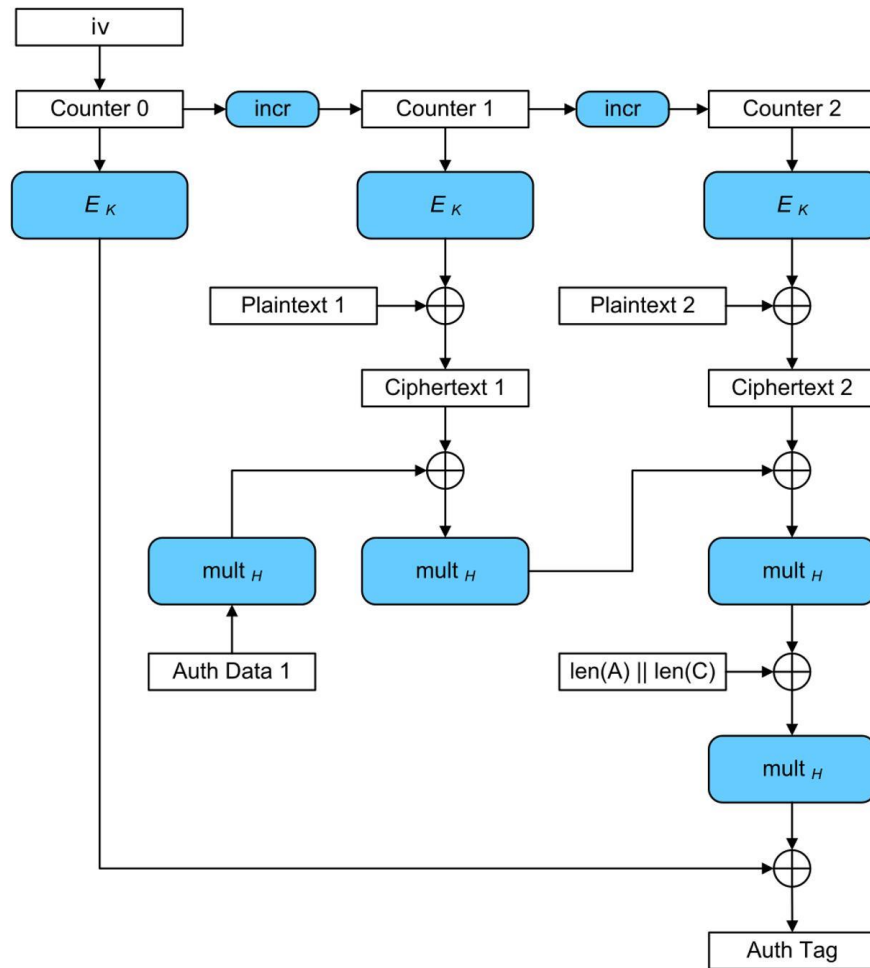
It provides cryptographic assurance that the ciphertext has not been altered by an unauthorized party possessing the required authentication information.

5. Suitable for modern systems

AES-GCM is designed for computer networks and digital communication and is much more appropriate for corporate systems than a classical transposition cipher.

Comparison: Legacy Method vs Modern Method

Feature	3-Rail Rail Fence	AES-GCM
Type	Transposition	Modern authenticated encryption
Changes letters?	No	Yes
Key complexity	Very low	High
Brute-force resistance	Very weak	Strong when properly configured
Integrity protection	No	Yes
Authentication	No	Yes
Modern communication	Not suitable	Suitable
Protection against modification	No	Yes
Suitable for confidential corporate data	No	Yes



Overall Conclusion for Question 5

The ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

can be interpreted as the result of a **3-rail Rail Fence transposition cipher**.

Reversing the transposition produces:

WE ARE DISCOVERED FLEE AT ONCE

Although the method can hide the message from someone who does not know the arrangement, it provides very weak security by modern standards. It preserves the original letters, has a small key space, can be attacked through trial and pattern analysis, and provides no authentication or integrity protection.

A practical improvement would therefore be to replace the legacy method with a modern authenticated encryption scheme such as **AES-GCM**, with proper key management and secure communication protocols.